

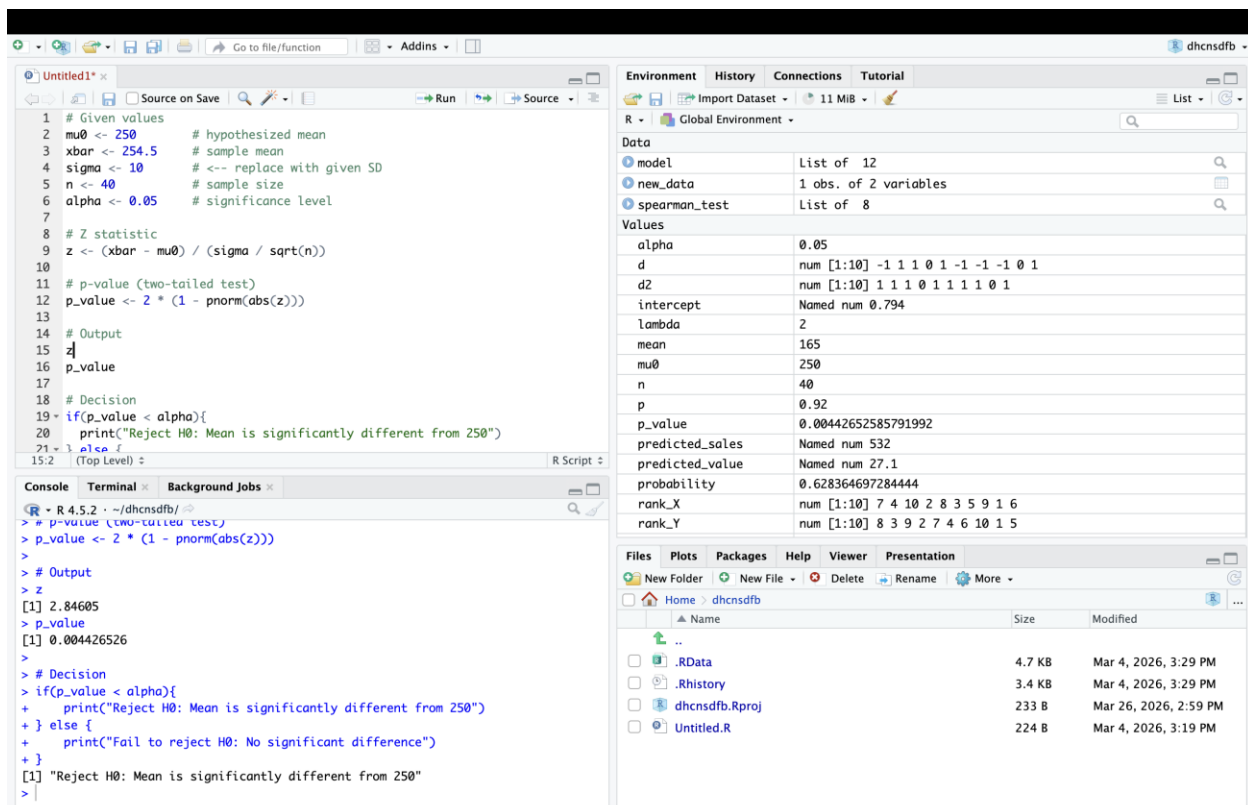
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Probability and Statistics Lab (DA-4)

1. A pharmaceutical manufacturer produces tablets that are supposed to contain an average of 250 mg of an active ingredient. The process is known to have a population standard deviation of σ mg. To ensure the machines are calibrated correctly, a quality control engineer selects a random sample of 40 tablets and finds the mean content to be 254.5 mg. At the α level of significance, is there sufficient evidence to conclude that the mean amount of the active ingredient is different from the claimed 250 mg?

CODE:



Que 1 (code part):-

Given values

mu0 <- 250 # hypothesized mean xbar <- 254.5 # sample mean sigma <- 10 # <-- replace with given SD n <- 40 # sample size alpha <- 0.05 # significance level

Z statistic

$z <- (xbar - mu0) / (sigma / \sqrt{n})$

p-value (two-tailed test)

$p_value <- 2 * (1 - pnorm(abs(z)))$

Output

z p_value

Decision

```
if(p_value < alpha){ print("Reject H0: Mean is significantly different from 250") } else  
{ print("Fail to reject H0: No significant difference") }
```

QUE-2

1. A new organic fertilizer is claimed to increase the yield of tomato plants. Without the fertilizer, the national average yield is 20 lbs per plant with a known population standard deviation of lbs. A researcher applies the new fertilizer to a random sample of 50 plants and records an average yield of 21.2 lbs per plant. At the level of significance, does the data support the claim that the new fertilizer significantly increases the average yield?

CODE:-

The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains an R script with the following code:


```

1 mu0 <- 20
2 xbar <- 21.2
3 sigma <- 4 # replace if given
4 n <- 50
5 alpha <- 0.05
6
7 z <- (xbar - mu0) / (sigma / sqrt(n))
8 p_value <- 1 - pnorm(z) # right-tailed
9
10 z
11 p_value
12
13 if(p_value < alpha){
14   print("Reject H0: Fertilizer increases yield")
15 } else {
16   print("Fail to reject H0")
17 }
18

```
- Console:** Shows the execution of the script:


```

> mu0 <- 20
>
> z <- (xbar - mu0) / (sigma / sqrt(n))
> p_value <- 1 - pnorm(z) # right-tailed
>
> z
[1] 2.12132
> p_value
[1] 0.01694743
>
> if(p_value < alpha){
+   print("Reject H0: Fertilizer increases yield")
+ } else {
+   print("Fail to reject H0")
+ }
[1] "Reject H0: Fertilizer increases yield"
>

```
- Environment Pane:** Lists the objects in the global environment:

Object	Type	Value
alpha	num	0.05
d	num [1:10]	-1 1 1 0 1 -1 -1 0 1
d2	num [1:10]	1 1 1 0 1 1 1 0 1
intercept	Named num	0.794
lambda	num	2
mean	num	165
mu0	num	20
n	num	50
p	num	0.92
p_value	num	0.0169474267623447
predicted_sales	Named num	532
predicted_value	Named num	27.1
probability	num	0.628364697284444
rank_X	num [1:10]	7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10]	8 3 9 2 7 4 6 10 1 5
- Files Pane:** Shows the file explorer with the following files:

Name	Size	Modified
..		
.RData	4.7 KB	Mar 4, 2026, 3:29 PM
.Rhistory	3.4 KB	Mar 4, 2026, 3:29 PM
dhcnsdfb.Rproj	233 B	Mar 26, 2026, 2:59 PM
Untitled.R	224 B	Mar 4, 2026, 3:19 PM

Que-2(code part):-

mu0 <- 20

xbar <- 21.2

sigma <- 4 # replace if given

n <- 50

alpha <- 0.05

z <- (xbar - mu0) / (sigma / sqrt(n))

p_value <- 1 - pnorm(z) # right-tailed

z

p_value

if(p_value < alpha){

```

print("Reject H0: Fertilizer increases yield")
} else {
  print("Fail to reject H0")
}

```

Que-3:-

1. A car manufacturer advertises that their new hybrid model achieves a mean fuel efficiency of 55 miles per gallon (mpg) on the highway. A consumer advocacy group suspects that the actual efficiency is lower than advertised. They know from independent testing that the population standard deviation is 6 mpg. They test a random sample of 32 cars and find a sample mean of 53.1 mpg. Test the group's suspicion at the 0.05 level of significance.

CODE:-

The screenshot displays the R Studio environment with a script editor on the left and the Environment pane on the right. The script performs a one-sample z-test for a mean.

```

1 mu0 <- 55
2 xbar <- 53.1
3 sigma <- 6 # replace if given
4 n <- 32
5 alpha <- 0.05
6
7 z <- (xbar - mu0) / (sigma / sqrt(n))
8 p_value <- pnorm(z) # left-tailed
9
10 z
11 p_value
12
13 if(p_value < alpha){
14   print("Reject H0: Efficiency is lower than 55 mpg")
15 } else {
16   print("Fail to reject H0")
17 }

```

The Environment pane shows the following objects and their values:

Object	Value
alpha	0.05
d	num [1:10] -1 1 1 0 1 -1 -1 0 1
d2	num [1:10] 1 1 1 0 1 1 1 1 0 1
intercept	Named num 0.794
lambda	2
mean	165
mu0	55
n	32
p	0.92
p_value	0.0366196016663125
predicted_sales	Named num 532
predicted_value	Named num 27.1
probability	0.628364697284444
rank_X	num [1:10] 7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10] 8 3 9 2 7 4 6 10 1 5

The Console shows the execution output:

```

> mu0 <- 55
> xbar <- 53.1
> sigma <- 6
> n <- 32
> alpha <- 0.05
>
> z <- (xbar - mu0) / (sigma / sqrt(n))
> p_value <- pnorm(z) # left-tailed
>
> z
[1] -1.791337
> p_value
[1] 0.0366196
>
> if(p_value < alpha){
+   print("Reject H0: Efficiency is lower than 55 mpg")
+ } else {
+   print("Fail to reject H0")
+ }
[1] "Reject H0: Efficiency is lower than 55 mpg"
>

```

Que-3(code part):-

```
mu0 <- 55
xbar <- 53.1
sigma <- 6 # replace if given
n <- 32
alpha <- 0.05

z <- (xbar - mu0) / (sigma / sqrt(n))
p_value <- pnorm(z) # left-tailed

z
p_value

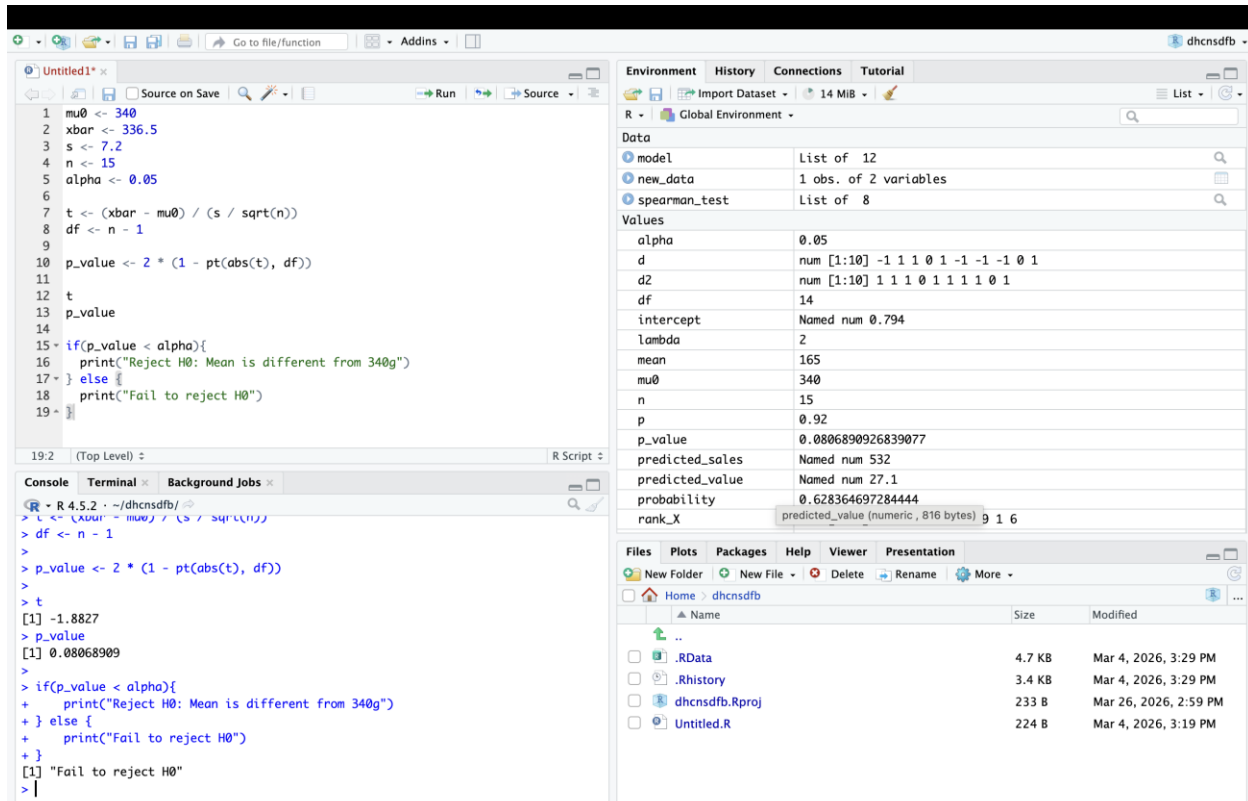
if(p_value < alpha){
  print("Reject H0: Efficiency is lower than 55 mpg")
} else {
  print("Fail to reject H0")
}
```

Que-4:-

1. A boutique coffee roaster claims that their "Signature Blend" bags contain an average of 340 g of coffee beans. A consumer advocate suspects the bags are not filled accurately and might be over or under the limit. They collect a random sample of 15 bags and find a sample

mean of 336.5 g with a sample standard deviation of 7.2 g. At the 5% significance level (), is there sufficient evidence to conclude that the actual mean weight is different from the claimed 340 g?

Code:-



The screenshot shows the RStudio environment with a script editor, console, and environment pane. The script performs a one-sample t-test. The console output shows the test statistic t = -1.8827 and the p-value = 0.08068909. Since the p-value is greater than the significance level alpha = 0.05, the conclusion is to fail to reject the null hypothesis.

```
1 mu0 <- 340
2 xbar <- 336.5
3 s <- 7.2
4 n <- 15
5 alpha <- 0.05
6
7 t <- (xbar - mu0) / (s / sqrt(n))
8 df <- n - 1
9
10 p_value <- 2 * (1 - pt(abs(t), df))
11
12 t
13 p_value
14
15 if(p_value < alpha){
16   print("Reject H0: Mean is different from 340g")
17 } else {
18   print("Fail to reject H0")
19 }
```

Console Output:

```
R > R 4.5.2 ~ /dhcnsdfb/
> t <- (xbar - mu0) / (s / sqrt(n))
> df <- n - 1
>
> p_value <- 2 * (1 - pt(abs(t), df))
>
> t
[1] -1.8827
> p_value
[1] 0.08068909
>
> if(p_value < alpha){
+   print("Reject H0: Mean is different from 340g")
+ } else {
+   print("Fail to reject H0")
+ }
[1] "Fail to reject H0"
> |
```

Environment Pane:

Object	Type	Value
alpha	num	0.05
d	num [1:10]	-1 1 1 0 1 -1 -1 -1 0 1
d2	num [1:10]	1 1 1 0 1 1 1 1 0 1
df	num	14
intercept	Named num	0.794
lambda	num	2
mean	num	165
mu0	num	340
n	num	15
p	num	0.92
p_value	num	0.0806890926839077
predicted_sales	Named num	532
predicted_value	Named num	27.1
probability	num	0.628364697284444
rank_X	predicted_value (numeric, 816 bytes)	9 1 6

Files Pane:

Name	Size	Modified
..	4.7 KB	Mar 4, 2026, 3:29 PM
.RData	3.4 KB	Mar 4, 2026, 3:29 PM
.Rhistory	233 B	Mar 26, 2026, 2:59 PM
dhcnsdfb.Rproj	224 B	Mar 4, 2026, 3:19 PM
Untitled.R		

Question-4(code):-

mu0 <- 340

xbar <- 336.5

s <- 7.2

n <- 15

```
alpha <- 0.05
```

```
t <- (xbar - mu0) / (s / sqrt(n))
```

```
df <- n - 1
```

```
p_value <- 2 * (1 - pt(abs(t), df))
```

```
t
```

```
p_value
```

```
if(p_value < alpha){
```

```
  print("Reject H0: Mean is different from 340g")
```

```
} else {
```

```
  print("Fail to reject H0")
```

```
}
```

Que-5:-

1. A software company claims that its new project management tool reduces the average time spent on weekly meetings, which was previously 120 minutes per team. A consultant believes the tool is effective and tests it with a random sample of 10 teams. After one month, the teams report a mean meeting time of 108 minutes with a sample standard deviation of 15 minutes. At the 1% significance level (), does the data support the claim that the tool significantly reduces the average meeting time?

Code:-

The screenshot shows the RStudio environment with the following components:

- Script Editor:** Contains R code for a hypothesis test.
- Console:** Shows the execution output of the script.
- Environment Pane:** Lists objects in the global environment.
- Files Pane:** Shows the file structure of the project.

```
1 mu0 <- 120
2 xbar <- 108
3 s <- 15
4 n <- 10
5 alpha <- 0.01
6
7 t <- (xbar - mu0) / (s / sqrt(n))
8 df <- n - 1
9
10 p_value <- pt(t, df) # left-tailed
11
12 t
13 p_value
14
15 if(p_value < alpha){
16   print("Reject H0: Tool reduces meeting time")
17 } else {
18   print("Fail to reject H0")
19 }
20
```

Console Output:

```
> t <- (xbar - mu0) / (s / sqrt(n))
> df <- n - 1
>
> p_value <- pt(t, df) # left-tailed
>
> t
[1] -2.529822
> p_value
[1] 0.01612239
>
> if(p_value < alpha){
+   print("Reject H0: Tool reduces meeting time")
+ } else {
+   print("Fail to reject H0")
+ }
[1] "Fail to reject H0"
>
```

Environment Pane:

Object	Type	Value
model	List of 12	
new_data	1 obs. of 2 variables	
spearman_test	List of 8	
alpha	0.01	
d	num [1:10]	-1 1 1 0 1 -1 -1 0 1
d2	num [1:10]	1 1 1 0 1 1 1 1 0 1
df	9	
intercept	Named num 0.794	
lambda	2	
mean	165	
mu0	120	
n	10	
p	0.92	
p_value	0.0161223904802284	
predicted_sales	Named num 532	
predicted_value	Named num 27.1	
probability	0.628364697284444	
rank_X	num [1:10]	7 4 10 2 8 3 5 9 1 6

Files Pane:

Name	Size	Modified
..		
.RData	4.7 KB	Mar 4, 2026, 3:29 PM
.Rhistory	3.4 KB	Mar 4, 2026, 3:29 PM
dhcnsdfb.Rproj	233 B	Mar 26, 2026, 2:59 PM
Untitled.R	224 B	Mar 4, 2026, 3:19 PM

Code-part:-

```
mu0 <- 120
```

```
xbar <- 108
```

```
s <- 15
```

```
n <- 10
```

```
alpha <- 0.01
```

```
t <- (xbar - mu0) / (s / sqrt(n))
```

```
df <- n - 1
```

```
p_value <- pt(t, df) # left-tailed
```

```
t
```

```
p_value
```

```
if(p_value < alpha){  
  print("Reject H0: Tool reduces meeting time")  
} else {  
  print("Fail to reject H0")  
}
```